

Most Significant Publications

Tejaswi Venumadhav Nerella

1. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030
Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo
2. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009
Title: New binary black hole mergers in the LIGO-Virgo O3a data
3. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), Physical Review D, 100, 023007
Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run
4. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011
Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO
5. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030
Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers

Refereed Publications

1. Islam, T., **Venumadhav, T.**, Wadekar, D., (2026), Physical Review Letters, 137, 031404
Title: Progenitor of the Recoiling Supermassive Black Hole RBH-1 Identified Using HST and JWST Imaging
2. Maity, K.N., Jana, S., **Venumadhav, T.**, Barsode, A., Ajith, P., (2026), Physical Review D, 113, 103522
Title: Strong lensing cosmography using binary-black-hole mergers: Prospects for the near future
3. Hegade K. R., A., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), Physical Review Letters, 136, 071401
Title: Relativistic and Dynamical Love Numbers
4. Cheung, M.H.Y., Wadekar, D., Mehta, A.K., Islam, T., Roulet, J., Berti, E., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2026), Physical Review D, 113, 023003
Title: Searching for intermediate mass ratio binary black hole mergers in the third observing run of LIGO-Virgo-KAGRA
5. Mehta, A.K., Wadekar, D., Anantpurkar, I., Roulet, J., **Venumadhav, T.**, Islam, T., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 124023
Title: Binary black hole population inference combining confident and marginal events from the IAS-HM search pipeline
6. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 112, 104039
Title: Post-Newtonian theory-inspired framework for characterizing eccentricity in gravitational waveforms
7. Perera, A., Zackay, B., **Venumadhav, T.**, (2025), Astrophys. J. Suppl. Ser., 281, 4
Title: A New Search Pipeline for Short Gamma-Ray Bursts in Fermi/GBM Data—A 50% Increase in the Number of Detections

8. Mushkin, J., Roulet, J., Zackay, B., **Venumadhav, T.**, Ivashtenko, O., Wadekar, D., Mehta, A.K., Zaldarriaga, M., (2025), Physical Review D, 112, 104025
Title: Sampler-free gravitational wave inference using matrix multiplication
9. Jana, S., Kapadia, S.J., **Venumadhav, T.**, More, S., Ajith, P., (2025), Physical Review Letters, 135, 111402
Title: Probing the Nature of Dark Matter Using Strongly Lensed Gravitational Waves from Binary Black Holes
10. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 112, 044070
Title: Data-driven extraction, phenomenology, and modeling of eccentric harmonics in binary black hole merger waveforms
11. Islam, T., **Venumadhav, T.**, (2025), Physical Review D, 111, L081503
Title: Universal phenomenological relations between spherical harmonic modes in nonprecessing eccentric binary black hole merger waveforms
12. Mehta, A.K., Olsen, S., Wadekar, D., Roulet, J., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), Physical Review D, 111, 024049
Title: New binary black hole mergers in the LIGO-Virgo O3b data
13. Jana, S., J Kapadia, S., **Venumadhav, T.**, More, S., Ajith, P., (2024), Classical and Quantum Gravity, 41, 245010
Title: Strong-lensing cosmography using third-generation gravitational-wave detectors
14. Wadekar, D., **Venumadhav, T.**, Mehta, A.K., Roulet, J., Olsen, S., Mushkin, J., Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 084035
Title: New approach to template banks of gravitational waves with higher harmonics: Reducing matched-filtering cost by over an order of magnitude
15. Chia, H.S., Edwards, T.D.P., Wadekar, D., Zimmerman, A., Olsen, S., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 063007
Title: In pursuit of Love numbers: First templated search for compact objects with large tidal deformabilities in the LIGO-Virgo data
16. Wadekar, D., **Venumadhav, T.**, Roulet, J., Mehta, A.K., Zackay, B., Mushkin, J., Zaldarriaga, M., (2024), Physical Review D, 110, 044063
Title: New search pipeline for gravitational waves with higher-order modes using mode-by-mode filtering
17. Roulet, J., Mushkin, J., Wadekar, D., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2024), Physical Review D, 110, 044010
Title: Fast marginalization algorithm for optimizing gravitational wave detection, parameter estimation, and sky localization
18. Roulet, J., **Venumadhav, T.**, (2024), Annu. Rev. Nucl. Part. Sci., 74, 207
Title: Inferring Binary Properties from Gravitational-Wave Signals
19. Yu, H., Roulet, J., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2023), Physical Review D, 108, 064059
Title: Accurate and efficient waveform model for precessing binary black holes
20. Jana, S., Kapadia, S.J., **Venumadhav, T.**, Ajith, P., (2023), Physical Review Letters, 130, 261401
Title: Cosmography Using Strongly Lensed Gravitational Waves from Binary Black Holes
21. Yu, H., Weinberg, N.N., Arras, P., Kwon, J., **Venumadhav, T.**, (2023), Mon. Not. R. Astron. Soc., 519, 4325

Title: Beyond the linear tide: impact of the non-linear tidal response of neutron stars on gravitational waveforms from binary inspirals

22. Roulet, J., Olsen, S., Mushkin, J., Islam, T., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 123015
Title: Removing degeneracy and multimodality in gravitational wave source parameters
23. Olsen, S., **Venumadhav, T.**, Mushkin, J., Roulet, J., Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 043009
Title: New binary black hole mergers in the LIGO-Virgo O3a data
24. Chia, H.S., Olsen, S., Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2022), Physical Review D, 106, 024009
Title: Signs of higher multipoles and orbital precession in GW151226
25. Olsen, S., Roulet, J., Chia, H.S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083036
Title: Mapping the likelihood of GW190521 with diverse mass and spin priors
26. Roulet, J., Chia, H.S., Olsen, S., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2021), Physical Review D, 104, 083010
Title: Distribution of effective spins and masses of binary black holes from the LIGO and Virgo O1-O3a observing runs
27. Zackay, B., **Venumadhav, T.**, Roulet, J., Dai, L., Zaldarriaga, M., (2021), Physical Review D, 104, 063034
Title: Detecting gravitational waves in data with non-stationary and non-Gaussian noise
28. Zackay, B., Dai, L., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2021), Physical Review D, 104, 063030
Title: Detecting gravitational waves with disparate detector responses: Two new binary black hole mergers
29. Roulet, J., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 123022
Title: Binary black hole mergers from LIGO/Virgo O1 and O2: Population inference combining confident and marginal events
30. Huang, Y., Haster, C.J., Roulet, J., Vitale, S., Zimmerman, A., **Venumadhav, T.**, Zackay, B., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 102, 103024
Title: Source properties of the lowest signal-to-noise-ratio binary black hole detections
31. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2020), Physical Review D, 101, 083030
Title: New binary black hole mergers in the second observing run of Advanced LIGO and Advanced Virgo
32. Dai, L., Kaurov, A.A., Sharon, K., Florian, M., Miralda-Escudé, J., **Venumadhav, T.**, Frye, B., Rigby, J.R., Bayliss, M., (2020), Mon. Not. R. Astron. Soc., 495, 3192
Title: Asymmetric surface brightness structure of caustic crossing arc in SDSS J1226+2152: a case for dark matter substructure
33. Samsing, J., **Venumadhav, T.**, Dai, L., Martinez, I., Batta, A., Lopez, M., Ramirez-Ruiz, E., Kremer, K., (2019), Physical Review D, 100, 043009
Title: Probing the black hole merger history in clusters using stellar tidal disruptions
34. **Venumadhav, T.**, Zackay, B., Roulet, J., Dai, L., Zaldarriaga, M., (2019), Physical Review D, 100, 023011

Title: New search pipeline for compact binary mergers: Results for binary black holes in the first observing run of Advanced LIGO

35. Kaurov, A.A., Dai, L., **Venumadhav, T.**, Miralda-Escudé, J., Frye, B., (2019), *Astrophysical Journal*, 880, 58
Title: Highly Magnified Stars in Lensing Clusters: New Evidence in a Galaxy Lensed by MACS J0416.1-2403
36. Zackay, B., **Venumadhav, T.**, Dai, L., Roulet, J., Zaldarriaga, M., (2019), *Physical Review D*, 100, 023007
Title: Highly spinning and aligned binary black hole merger in the Advanced LIGO first observing run
37. Roulet, J., Dai, L., **Venumadhav, T.**, Zackay, B., Zaldarriaga, M., (2019), *Physical Review D*, 99, 123022
Title: Template bank for compact binary coalescence searches in gravitational wave data: A general geometric placement algorithm
38. **Venumadhav, T.**, Dai, L., Kaurov, A., Zaldarriaga, M., (2018), *Physical Review D*, 98, 103513
Title: Heating of the intergalactic medium by the cosmic microwave background during cosmic dawn
39. Dai, L., **Venumadhav, T.**, Kaurov, A.A., Miralda-Escudé, J., (2018), *Astrophysical Journal*, 867, 24
Title: Probing Dark Matter Subhalos in Galaxy Clusters Using Highly Magnified Stars
40. Kaurov, A.A., **Venumadhav, T.**, Dai, L., Zaldarriaga, M., (2018), *Astrophys. J. Lett.*, 864, L15
Title: Implication of the Shape of the EDGES Signal for the 21 cm Power Spectrum
41. Hirata, C.M., Mishra, A., **Venumadhav, T.**, (2018), *Physical Review D*, 97, 103521
Title: Detecting primordial gravitational waves with circular polarization of the redshifted 21 cm line. I. Formalism
42. **Venumadhav, T.**, Dai, L., Miralda-Escudé, J., (2017), *Astrophysical Journal*, 850, 49
Title: Microlensing of Extremely Magnified Stars near Caustics of Galaxy Clusters
43. **Venumadhav, T.**, Oklopčić, A., Gluscevic, V., Mishra, A., Hirata, C.M., (2017), *Physical Review D*, 95, 083010
Title: New probe of magnetic fields in the preionization epoch. I. Formalism
44. Gluscevic, V., **Venumadhav, T.**, Fang, X., Hirata, C., Oklopčić, A., Mishra, A., (2017), *Physical Review D*, 95, 083011
Title: New probe of magnetic fields in the pre-reionization epoch. II. Detectability
45. Dai, L., **Venumadhav, T.**, Sigurdson, K., (2017), *Physical Review D*, 95, 044011
Title: Effect of lensing magnification on the apparent distribution of black hole mergers
46. **Venumadhav, T.**, Chang, T.C., Doré, O., Hirata, C.M., (2016), *Astrophysical Journal*, 826, 116
Title: A Practical Theorem on Using Interferometry to Measure the Global 21-cm Signal
47. **Venumadhav, T.**, Cyr-Racine, F.Y., Abazajian, K.N., Hirata, C.M., (2016), *Physical Review D*, 94, 043515
Title: Sterile neutrino dark matter: Weak interactions in the strong coupling epoch
48. **Venumadhav, T.**, Hirata, C., (2015), *Physical Review D*, 91, 123009
Title: Stability of small-scale baryon perturbations during cosmological recombination
49. **Venumadhav, T.**, Zimmerman, A., Hirata, C.M., (2014), *Astrophysical Journal*, 781, 23
Title: The Stability of Tidally Deformed Neutron Stars to Three- and Four-mode Coupling

50. Venumadhav, T., Haque, M., Moessner, R., (2010), Physical Review B, 81, 054305

Title: Finite-rate quenches of site bias in the Bose-Hubbard dimer

Preprints

1. Ho-Yeuk Cheung, M., Wadekar, D., Zaldarriaga, M., **Venumadhav, T.**, Roulet, J., Mehta, A.K., (2026), arXiv:2607.21834

Title: The diffraction-lensing interpretation of GW231123 with astrophysical priors

2. Yu, H., Nicolini, G., Lau, S.Y., Kwon, K.J., **Venumadhav, T.**, Andersson, N., Pnigouras, P., Gittins, F., Nanda, A., (2026), arXiv:2607.07943

Title: Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions

3. Abhishek Hegade K., R., Kwon, K.J., **Venumadhav, T.**, Yu, H., Yunes, N., (2026), arXiv:2605.08569

Title: The Good, the Bad, and the Subtle: Relativistic mode sums for neutron-star tidal response

4. Perera, A., Zackay, B., **Venumadhav, T.**, (2026), arXiv:2605.31554

Title: Expanding the Population of Short Gamma-Ray Transients with a Coherent Fermi/GBM Search. A 13-year catalog of short GRBs

5. Islam, T., Wadekar, D., **Venumadhav, T.**, Zaldarriaga, M., Mehta, A.K., Roulet, J., Zackay, B., (2026), arXiv:2605.11280

Title: Discovery of Interpretable Surrogates via Agentic AI: Application to Gravitational Waves

6. Islam, T., **Venumadhav, T.**, Wadekar, D., Mehta, A.K., Roulet, J., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2604.07388

Title: GW190711_030756 and GW200114_020818: astrophysical interpretation of two asymmetric binary black hole mergers in the IAS catalog

7. Zhou, Z., Wadekar, D., Roulet, J., Ivashtenko, O., **Venumadhav, T.**, Islam, T., Mehta, A.K., Mushkin, J., Ho-Yeuk Cheung, M., Zackay, B., Zaldarriaga, M., (2026), arXiv:2603.05784

Title: Searching for precessing binary systems with mode-by-mode filtering and marginalization

8. Islam, T., **Venumadhav, T.**, Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2509.20556

Title: Data-driven extraction and phenomenology of eccentric harmonics in eccentric spinning binary black hole mergers

9. Wadekar, D., Pimpalkar, A., Ho-Yeuk Cheung, M., Wandelt, B., Berti, E., Mehta, A.K., **Venumadhav, T.**, Roulet, J., Islam, T., Zackay, B., Mushkin, J., Zaldarriaga, M., (2025), arXiv:2507.08318

Title: Improving gravitational wave search sensitivity with TIER: Trigger Inference using Extended strain Representation

10. Schiff, J., **Venumadhav, T.**, (2025), arXiv:2506.16517

Title: Primordial magnetic fields and modified recombination histories

11. Islam, T., **Venumadhav, T.**, Mehta, A.K., Anantpurkar, I., Wadekar, D., Roulet, J., Mushkin, J., Zackay, B., Zaldarriaga, M., (2025), arXiv:2504.12420

Title: gwharmone: first data-driven surrogate for eccentric harmonics in binary black hole merger waveforms

12. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2025), arXiv:2503.11837

Title: Resonance locking: radian-level phase shifts due to nonlinear hydrodynamics of g -modes in merging neutron star binaries

13. Mehta, A.K., Wadekar, D., Roulet, J., Anantpurkar, I., **Venumadhav, T.**, Mushkin, J., Zackay, B., Zaldarriaga, M., Islam, T., (2025), arXiv:2501.17939
Title: Significant increase in sensitive volume of a gravitational wave search upon including higher harmonics
14. Kwon, K.J., Yu, H., **Venumadhav, T.**, (2024), arXiv:2410.03831
Title: Resonance Locking of Anharmonic g -Modes in Coalescing Neutron Star Binaries
15. Wadekar, D., Roulet, J., **Venumadhav, T.**, Mehta, A.K., Zackay, B., Mushkin, J., Olsen, S., Zaldarriaga, M., (2023), arXiv:2312.06631
Title: New black hole mergers in the LIGO-Virgo O3 data from a gravitational wave search including higher-order harmonics
16. Islam, T., Roulet, J., **Venumadhav, T.**, (2022), arXiv:2210.16278
Title: Factorized Parameter Estimation for Real-Time Gravitational Wave Inference
17. Dai, L., Zackay, B., **Venumadhav, T.**, Roulet, J., Zaldarriaga, M., (2020), arXiv:2007.12709
Title: Search for Lensed Gravitational Waves Including Morse Phase Information: An Intriguing Candidate in O2
18. Coleman, M.S.B., **Venumadhav, T.**, Zackay, B., (2019), arXiv:1903.04978
Title: Gravitational-wave-moderated Accretion: The Case of ES Ceti
19. Haris, K., Mehta, A.K., Kumar, S., **Venumadhav, T.**, Ajith, P., (2018), arXiv:1807.07062
Title: Identifying strongly lensed gravitational wave signals from binary black hole mergers
20. Zackay, B., Dai, L., **Venumadhav, T.**, (2018), arXiv:1806.08792
Title: Relative Binning and Fast Likelihood Evaluation for Gravitational Wave Parameter Estimation
21. Dai, L., **Venumadhav, T.**, Zackay, B., (2018), arXiv:1806.08793
Title: Parameter Estimation for GW170817 using Relative Binning
22. Dai, L., **Venumadhav, T.**, (2017), arXiv:1702.04724
Title: On the waveforms of gravitationally lensed gravitational waves

n^{th} Author Papers

1. Raaijmakers, G., et. al., (2021), Astrophysical Journal, 922, 269
Title: The Challenges Ahead for Multimessenger Analyses of Gravitational Waves and Kilonova: A Case Study on GW190425
2. Weltman, A., et. al., (2020), Publ. Astron. Soc. Aust., 37, e002
Title: Fundamental physics with the Square Kilometre Array
3. Doré, O., et. al., (2014), arXiv:1412.4872
Title: Cosmology with the SPHEREX All-Sky Spectral Survey